initial_findings_report_22_09_2025

Project Mycelium -- Progress Report #1

```
Author(s): Shasank Prasad
```

Initial Findings

September 22, 2025

As previously discussed, we agreed that I would start the initial work on Project Mycelium, particularly get layer 1 of the project working and see how it works, whether it lines up with initial planning and expectations.

As of 12:00 PM currently I am quite content to say that initial progress has been satisfactory, and quite interesting. The architecture accurately generates tags based on the current input (I have decided to keep the input type to be a string sentence for our early working demo, we can test out with other input types like images or videos or a mix of inputs later down the line), where each tag corresponds to a domain of a specific topic, like artificial intelligence, healthcare, finance etc. The layer then proceeds to cluster all the tags using agglomerative clustering to group all the tags together which seem to have close relations to each other.

The first layer also assigns timestamps to input statements and maintains a like queue data structure to store the inputs in order of their timestamps, this, along with the clustering will help with the cold storage layer later down the line.

The second layer, for now, simulates models with their simulated metric scores like precision, recall, F1 scores, MSE, MAE scores, to test out whether the decision making layer of :

- Creating a new expert
- Choosing a hybrid of experts from neighbouring tags in a single cluster.
- Creating a new patch network
- Choosing a hybrid of patch networks from neighbouring tags in a single cluster,

works or not. For now, with the evaluation report, it seems that the decision making layer does in fact, work, and can be further improved down the line, when real models are plugged in with more efficient scoring algorithms to judge the competency of the domain expert/patch or both.

How the prototype works so far.

Layer 1

```
1. New input arrives.



2. A pre-prompted LLM is tasked to analyze the input and generate the tags
based on where the input leans towards. For simple sentence based inputs we
can just use a small language model as low as 3b or 4b parameter type models
(llama3.2, phi3). For this particular test however an, 8b version of
llama3.1 was used.



3. The tags are then taken and converted to vector embeddings and put
through a powerful transformer model `all-mpnet-base-v2` and then put
through agglomerative clustering with a threshold of 0.5.



4. Next the existing experts per domains are assesed using their metric
scores for now, to check which domain expert can cater well to the current
input. This particular part will be improved next with much better
algorithms and with real models instead of simulated ones.



5. The generated information is stored in separate JSON files as follows:
```

`Sentence Tags and Experts information:`

```
[
    {
        "sentence": "AI-driven systems can analyze data, improve healthcare, and optimize logistics",
        "tags": [
            "AI",
            "healthcare",
            "logistics"
        ],
        "timestamp": "2025-09-21T23:36:41.817631",
        "expert_scores": {
            "AI": {
                "precision": 0.76,
                "recall": 0.88,
                "f1": 0.82,
                "mse": 0.09,
                "mae": 0.02
            },
            "healthcare": {
                "precision": 0.92,
                "recall": 0.89,
                "f1": 0.9,
```

```
                "mse": 0.09,

                "mae": 0.04

            }

        },

        "expert_flag": "create_hybrid_expert"

    },

    {

        "sentence": "Quantum computers are revolutionizing physics and
mathematics research",

        "tags": [

            "AI",

            "physics",

            "maths"

        ],

        "timestamp": "2025-09-21T23:36:42.061426",

        "expert_scores": {

            "AI": {

                "precision": 0.91,

                "recall": 0.84,

                "f1": 0.87,

                "mse": 0.06,

                "mae": 0.08
```

```
            },
            "physics": {
                "precision": 0.92,
                "recall": 0.83,
                "f1": 0.87,
                "mse": 0.05,
                "mae": 0.06
            }
        },
        "expert_flag": "create_hybrid_expert"
    },
    {
        "sentence": "Financial markets are influenced by global politics and economic trends",
        "tags": [
            "finance",
            "politics"
        ],
        "timestamp": "2025-09-21T23:36:42.230646",
        "expert_scores": {
            "finance": {
                "precision": 0.89,
```

```
                "recall": 0.84,

                "f1": 0.86,

                "mse": 0.05,

                "mae": 0.04

            }

        },

        "expert_flag": "use_existing_expert"

    },

    {

        "sentence": "Music and art therapy are used in modern healthcare for
mental wellness",

        "tags": [

            "healthcare",

            "art",

            "music"

        ],

        "timestamp": "2025-09-21T23:36:42.456050",

        "expert_scores": {

            "healthcare": {

                "precision": 0.93,

                "recall": 0.83,

                "f1": 0.88,
```

```
                "mse": 0.03,

                "mae": 0.02

            },

            "music": {

                "precision": 0.75,

                "recall": 0.72,

                "f1": 0.73,

                "mse": 0.1,

                "mae": 0.02

            }

        },

        "expert_flag": "create_hybrid_expert"

    },

    {

        "sentence": "Education technology platforms leverage AI to
personalize learning",

        "tags": [

            "AI",

            "education",

            "technology"

        ],

        "timestamp": "2025-09-21T23:36:42.660542",
```

```
        "expert_scores": {

            "AI": {

                "precision": 0.73,

                "recall": 0.72,

                "f1": 0.72,

                "mse": 0.03,

                "mae": 0.06

            }

        },

        "expert_flag": "use_existing_expert"

    },

    {

        "sentence": "Sports analytics use big data to optimize team
performance",

        "tags": [

            "sports",

            "technology"

        ],

        "timestamp": "2025-09-21T23:36:42.830993",

        "expert_scores": {},

        "expert_flag": "create_new_expert"

    },
```

```
    {
        "sentence": "Biology and chemistry are foundational for
pharmaceutical innovations",
        "tags": [
            "biology",
            "chemistry"
        ],
        "timestamp": "2025-09-21T23:36:43.000499",
        "expert_scores": {},
        "expert_flag": "create_new_expert"
    },
    {
        "sentence": "History and literature provide context for
understanding political movements",
        "tags": [
            "history",
            "literature",
            "politics"
        ],
        "timestamp": "2025-09-21T23:36:43.236273",
        "expert_scores": {},
        "expert_flag": "create_new_expert"
```

```
    },
    {
        "sentence": "Mathematics is essential for advancements in physics
and finance",
        "tags": [
            "maths",
            "physics",
            "finance"
        ],
        "timestamp": "2025-09-21T23:36:43.457198",
        "expert_scores": {
            "physics": {
                "precision": 0.8,
                "recall": 0.74,
                "f1": 0.77,
                "mse": 0.04,
                "mae": 0.06
            },
            "finance": {
                "precision": 0.73,
                "recall": 0.93,
                "f1": 0.82,
```

```
                "mse": 0.09,

                "mae": 0.09

            }

        },

        "expert_flag": "create_hybrid_expert"

    },

    {

        "sentence": "Logistics companies use AI and IoT to streamline supply
chains",

        "tags": [

            "AI",

            "logistics",

            "technology"

        ],

        "timestamp": "2025-09-21T23:36:43.659098",

        "expert_scores": {

            "AI": {

                "precision": 0.79,

                "recall": 0.74,

                "f1": 0.76,

                "mse": 0.05,

                "mae": 0.02
```

```
            }
        },
        "expert_flag": "use_existing_expert"
    }
]
```


`Clustering data`:


```
{
    "tags": [
        "art",
        "physics",
        "education",
        "politics",
        "healthcare",
        "AI",
        "maths",
        "biology",
        "literature",
        "technology",
        "music",
        "sports",
        "finance",
```

```
        "chemistry",
        "logistics",
        "history"
    ],
    "clusters": {
        "0": [
            "art",
            "literature",
            "music"
        ],
        "1": [
            "physics",
            "maths",
            "chemistry"
        ],
        "2": [
            "education"
        ],
        "3": [
            "politics"
        ],
```

```
"4": [
    "healthcare"
],
"5": [
    "AI"
],
"6": [
    "biology"
],
"7": [
    "technology"
],
"8": [
    "sports"
],
"9": [
    "finance",
    "logistics"
],
"10": [
    "history"
]
```

```
    }
}
```

> Note how meaningful clusters are formed, such as art, literature and music clustered together, same for physics, maths and chemistry and for finance and logistics as well.

`Expert evaluation results` (subject to be changed when real models with real data and better algorithms are used):


```json
{
    "evaluation_timestamp": "2025-09-21T23:36:43.660097",
    "sentences_evaluated": 10,
    "flag_summary": {
        "create_hybrid_expert": 4,
        "use_existing_expert": 3,
        "create_new_expert": 3
    },
    "detailed_results": [
        {
            "sentence": "AI-driven systems can analyze data, improve healthcare, and optimize logistics",
            "tags": [
                "AI",
                "healthcare",
                "logistics"
```

```
    ],
    "timestamp": "2025-09-21T23:36:41.817631",
    "expert_scores": {
        "AI": {
            "precision": 0.76,
            "recall": 0.88,
            "f1": 0.82,
            "mse": 0.09,
            "mae": 0.02
        },
        "healthcare": {
            "precision": 0.92,
            "recall": 0.89,
            "f1": 0.9,
            "mse": 0.09,
            "mae": 0.04
        }
    },
    "expert_flag": "create_hybrid_expert"
},
{
```

```
            "sentence": "Quantum computers are revolutionizing physics and
mathematics research",

            "tags": [

                "AI",

                "physics",

                "maths"

            ],

            "timestamp": "2025-09-21T23:36:42.061426",

            "expert_scores": {

                "AI": {

                    "precision": 0.91,

                    "recall": 0.84,

                    "f1": 0.87,

                    "mse": 0.06,

                    "mae": 0.08

                },

                "physics": {

                    "precision": 0.92,

                    "recall": 0.83,

                    "f1": 0.87,

                    "mse": 0.05,

                    "mae": 0.06
```

```
                }
            },
            "expert_flag": "create_hybrid_expert"
        },
        {
            "sentence": "Financial markets are influenced by global politics
and economic trends",
            "tags": [
                "finance",
                "politics"
            ],
            "timestamp": "2025-09-21T23:36:42.230646",
            "expert_scores": {
                "finance": {
                    "precision": 0.89,
                    "recall": 0.84,
                    "f1": 0.86,
                    "mse": 0.05,
                    "mae": 0.04
                }
            },
            "expert_flag": "use_existing_expert"
```

```
        },
        {
            "sentence": "Music and art therapy are used in modern healthcare for mental wellness",
            "tags": [
                "healthcare",
                "art",
                "music"
            ],
            "timestamp": "2025-09-21T23:36:42.456050",
            "expert_scores": {
                "healthcare": {
                    "precision": 0.93,
                    "recall": 0.83,
                    "f1": 0.88,
                    "mse": 0.03,
                    "mae": 0.02
                },
                "music": {
                    "precision": 0.75,
                    "recall": 0.72,
                    "f1": 0.73,
```

```
                    "mse": 0.1,
                    "mae": 0.02
                }
            },
            "expert_flag": "create_hybrid_expert"
        },
        {
            "sentence": "Education technology platforms leverage AI to
personalize learning",
            "tags": [
                "AI",
                "education",
                "technology"
            ],
            "timestamp": "2025-09-21T23:36:42.660542",
            "expert_scores": {
                "AI": {
                    "precision": 0.73,
                    "recall": 0.72,
                    "f1": 0.72,
                    "mse": 0.03,
                    "mae": 0.06
```

```
                }
            },
            "expert_flag": "use_existing_expert"
        },
        {
            "sentence": "Sports analytics use big data to optimize team performance",
            "tags": [
                "sports",
                "technology"
            ],
            "timestamp": "2025-09-21T23:36:42.830993",
            "expert_scores": {},
            "expert_flag": "create_new_expert"
        },
        {
            "sentence": "Biology and chemistry are foundational for pharmaceutical innovations",
            "tags": [
                "biology",
                "chemistry"
            ],
            "timestamp": "2025-09-21T23:36:43.000499",
```

```
            "expert_scores": {},
            "expert_flag": "create_new_expert"
        },
        {
            "sentence": "History and literature provide context for
understanding political movements",
            "tags": [
                "history",
                "literature",
                "politics"
            ],
            "timestamp": "2025-09-21T23:36:43.236273",
            "expert_scores": {},
            "expert_flag": "create_new_expert"
        },
        {
            "sentence": "Mathematics is essential for advancements in
physics and finance",
            "tags": [
                "maths",
                "physics",
                "finance"
```

```
    ],
    "timestamp": "2025-09-21T23:36:43.457198",
    "expert_scores": {
        "physics": {
            "precision": 0.8,
            "recall": 0.74,
            "f1": 0.77,
            "mse": 0.04,
            "mae": 0.06
        },
        "finance": {
            "precision": 0.73,
            "recall": 0.93,
            "f1": 0.82,
            "mse": 0.09,
            "mae": 0.09
        }
    },
    "expert_flag": "create_hybrid_expert"
},
{
    "sentence": "Logistics companies use AI and IoT to streamline
```

```
supply chains",
            "tags": [
                "AI",
                "logistics",
                "technology"
            ],
            "timestamp": "2025-09-21T23:36:43.659098",
            "expert_scores": {
                "AI": {
                    "precision": 0.79,
                    "recall": 0.74,
                    "f1": 0.76,
                    "mse": 0.05,
                    "mae": 0.02
                }
            },
            "expert_flag": "use_existing_expert"
        }
    ]
}
```

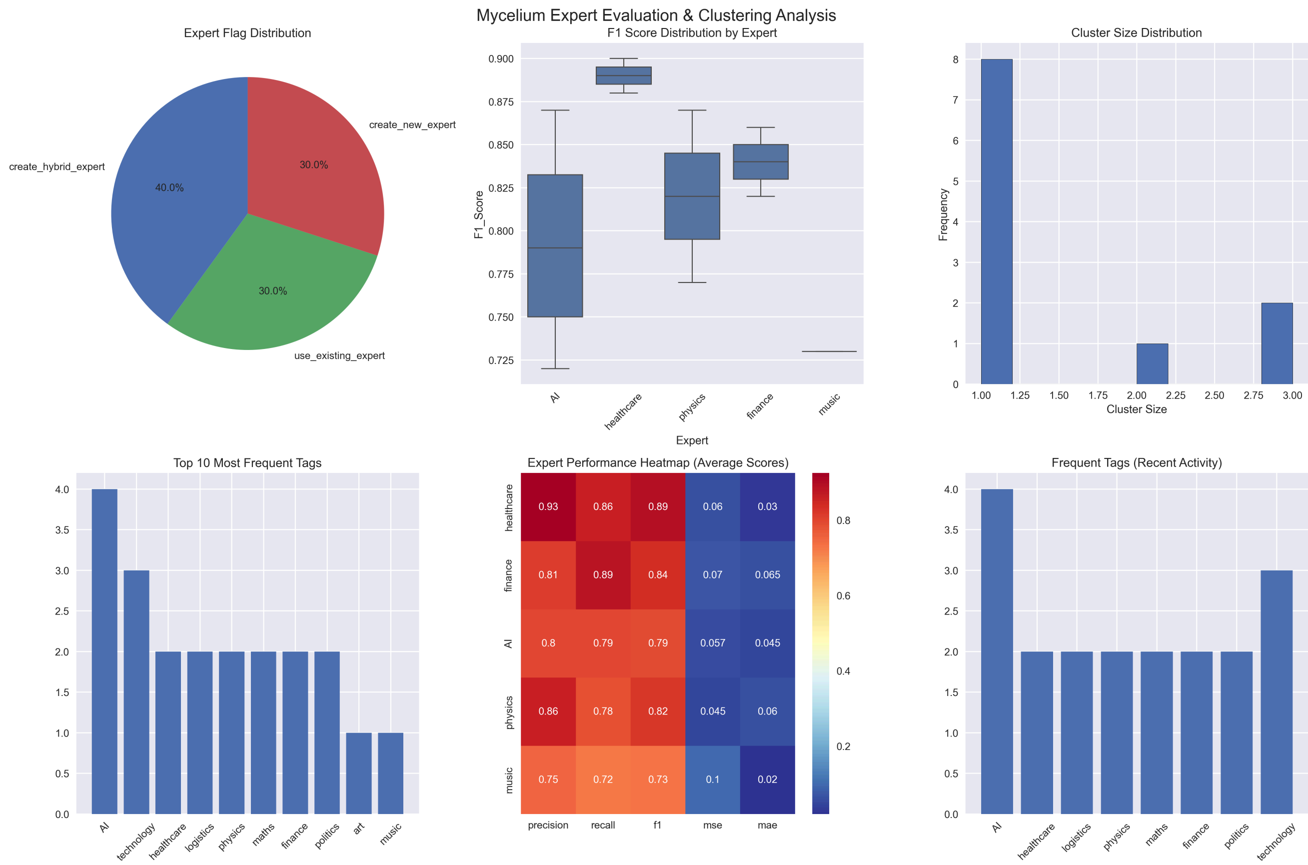

`Temporal analysis data` (will continue to change based on incoming inputs)

```
{
    "recent_statements_count": 10,
    "spatial_analysis": {
        "dominant_clusters": [
            [
                "1",
                5
            ],
            [
                "5",
                4
            ],
            [
                "9",
                4
            ]
        ],
        "cluster_distribution": {
            "5": 4,
            "4": 2,
            "9": 4,
```

```
            "1": 5,
            "3": 2,
            "0": 3,
            "2": 1,
            "7": 3,
            "8": 1,
            "6": 1,
            "10": 1
        },
        "total_recent_tags": 27,
        "unique_recent_tags": 16
    },
    "patch_assignment": {
        "action": "create_hybrid_patch",
        "primary_cluster": "1",
        "dominance_ratio": 0.18518518518518517,
        "reason": "moderate_dominance_0.19_suggests_hybrid"
    },
    "frequent_tags": {
        "AI": 4,
        "healthcare": 2,
        "logistics": 2,
```

```
            "physics": 2,
            "maths": 2,
            "finance": 2,
            "politics": 2,
            "technology": 3
        },
        "analysis_timestamp": "2025-09-21T23:36:50.930220"
    }
```

A visual representation of the above information



Final Thoughts

So, far work on this project has been really promising and satisfactory, enough that we can proceed into the next section of plugging in small models trained on real data for a few of the existing domain tags and plug in real evaluation algorithms than just doing basic threshold based checking.

Information on the next layer and how it can be potentially implemented will be discussed on in the next meeting session.
